LVDT displacement calibration

EMAE 285 · Lab 1

Experiment overview

Calibrated a linear variable differential transformer using known displacements and voltage output. Compared data-acquisition and multimeter readings, evaluated repeatability, linear operating range and departures from linearity.

Variables and instrumentation

Known displacement, transducer voltage, instrument-to-instrument consistency

Analysis and presentation

Calibration plot; measurement error; sensitivity and usable linear range

Engineering relevance

In automated inspection, a sensor is only useful if its output is calibrated and its uncertainty understood.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1JGTVENMrCybQ1Tam4GWjyiMaiw2bq5pZlc-F85dzRxc/edit